INCH-POUND

MIL-DTL-21097/2F
w/AMENDMENT 1
1 February 2017
SUPERSEDING
MIL-DTL-21097/2F
18 August 2005

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, PRINTED WIRING BOARD COMPOSITE, RECEPTACLE, CONTACT SPACING (.200) SINGLE ROW, TYPE CR

Inactive for new design after 29 March 1976. For new design, use MIL-DTL-21097/4.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-21097.

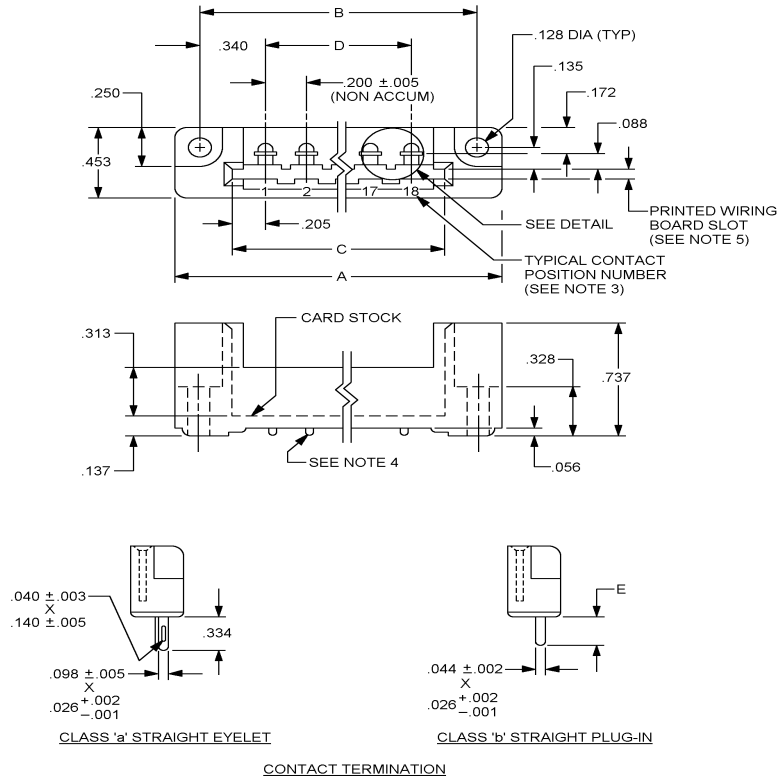
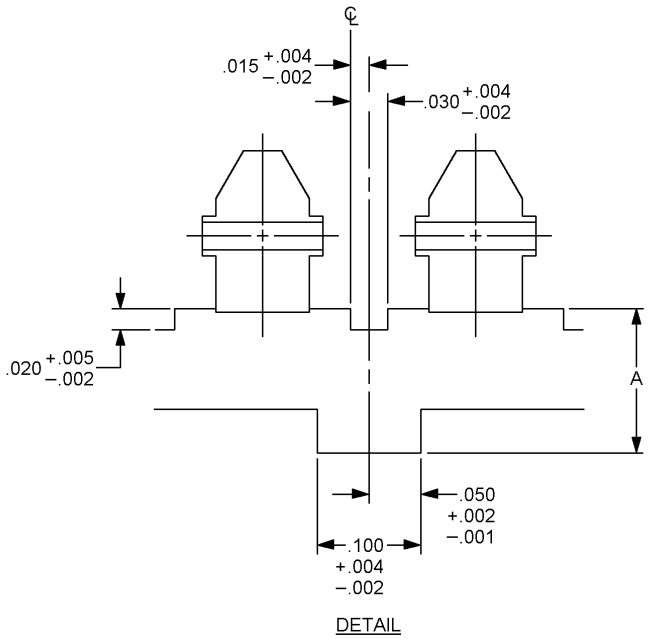


FIGURE 1. Single row contacts, type CR.

AMSC N/A

FSC 5935

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POLARIZING CAVITY FOR EVEN NUMBER OF CONTACT POSITIONS

Inches	mm	Inches	mm
.001	0.03	.100	2.54
.002	0.05	.128	3.25
.003	0.08	.135	3.43
.004	0.10	.137	3.48
.005	0.13	.140	3.56
.015	0.38	.172	4.37
.020	0.51	.200	5.08
.026	0.66	.205	5.21
.030	0.76	.250	6.35
.040	1.02	.313	7.95
.044	1.12	.326	8.28
.050	1.27	.334	8.48
.056	1.42	.340	8.64
.088	2.24	.453	11.51
.098	2.49	.737	18.72

Board guide	A dim + .002 (.05) - .004 (.10)
.06 (1.52) .09 (2.29)	.119 (3.02) .150 (3.81)

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .010 (.25 mm) for three place decimals.
3. Contacts and terminals shall be designated on the connector using numerals in the locations shown.
4. The shape of the barriers is optional except that barriers shall not protrude beyond the plane of the underside of the mounting flange.
5. Slot shall accept the following thickness printed wiring boards:
   Size 2 - .054 (1.37 mm) to .071 (1.80 mm)
   Size 3 - .086 (2.18 mm) to .103 (2.62 mm)
   Slot shall prevent entry of printed wiring board having the following thickness:
   Size 2 - .084 (2.13 mm)
   Size 3 - .113 (2.87 mm)
6. Metric equivalents are given for information only.
7. Millimeters are in parentheses.

FIGURE 1. Single row contacts, type CR – Continued.

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TABLE I. Connector, electrical, composite, receptacle, type CR, mounting provision A.

Dash number	Style	Class	Size	Dimensions (inches) 1/ 2/				
				A (Maximum)	B ±0.008 (0.20)	C +0.010 (0.25) -0.000	D ±0.005 (0.13)	E ±0.010 (0.25)
01		a	2					
02		a	3					
03		b	2					.123 (3.12)
04	12	b	2	3.150 (80.01)	2.880 (73.15)	2.610 (66.29)	2.200 (55.88)	.154 (3.91)
05		b	2					.186 (4.72)
06		b	3					.123 (3.12)
07		b	3					.154 (3.91)
08		b	3					.186 (4.72)
09		a	2					
10		a	3					
11		b	2					.123 (3.12)
12	15	b	2	3.750 (95.25)	3.480 (88.39)	3.210 (81.53)	2.800 (71.12)	.154 (3.91)
13		b	2					.186 (4.72)
14		b	3					.123 (3.12)
15		b	3					.154 (3.91)
16		b	3					.186 (4.72)

See footnotes at end of table.

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TABLE I. Connector, electrical, composite, receptacle, type CR, mounting provision A – Continued

Dash number	Style	Class	Size	Dimensions (inches) 1/ 2/ A (Maximum)	B ±0.008 (0.20)	C +0.010 -0.000	D ±0.005 (0.13)	E ±0.010 (0.25)
17		a	2					
18		a	3					.123
19		b	2					(3.12)
20	18	b	2	4.350 (110.49)	4.080 (103.63)	3.810 (96.77)	3.400 (86.36)	.154 (3.91)
21		b	2					.186 (4.72)
22		b	3					.123 (3.12)
23		b	3					.154 (3.91)
24		b	3					.186 (4.72)
25		a	2					
26		a	3					.123
27		b	2					(3.12)
28	22	b	2	5.150 (130.81)	4.880 (123.95)	4.610 (117.09)	4.200 (106.68)	.154 (3.91)
29		b	2					.186 (4.72)
30		b	3					.123 (3.12)
31		b	3					.154 (3.91)
32		b	3					.186 (4.72)
33		a	2					
34		a	3					
35		b	2					.123 (3.12)
36	26	b	2	5.950 (151.13)	5.680 (144.27)	5.410 (137.41)	5.000 (127.00)	.154 (3.91)
37		b	2					.186 (4.72)
38		b	3					.123 (3.12)
39		b	3					.154 (3.91)
40		b	3					.186 (4.72)

1/. Dimensions are in inches
2/. Metric equivalents are given for information only.

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REQUIREMENTS

Dimensions and configurations: See figure 1 and table I.

Polarizing keys: Part or identifying number (PIN) M21097/11-03 or M21097/11-04. A key is inserted into the card slot between contact numbers 1 and 2, 2 and 3, 3 and 4, etc.

Mating male adapters: Shall be in accordance with MIL-DTL-21097/3.

Finish: Gold plating in accordance with MIL-DTL-45204, type II, grade C, class 1, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 30 to 150 microinches.

Retention of qualification: Group A inspection only.

(PIN): M21097/2-(dash number from table I).

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-21097, this document references the following:

MIL-DTL-21097/3
MIL-DTL-45204
SAE-AMS-QQ-N-290

CONCLUDING MATERIALS

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2017-007)

Review activities:
Army – AR, AT, AV, MI
Navy – AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .

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